Ans. (a) : Cornea, the main focusing part, is the clear front surface of the eye. Like a window, it allows light to enter the eye. Vision could be markedly reduced or lost if the cornea becomes cloudy or scarred. This condition is known as corneal blindness.

Eye donation is an act of donating one's eyes after his/her death. Only corneal blinds can be benefited through this process not other blinds.

Cornea transplant is the surgical procedure which replaces a disc-shaped segment of an impaired cornea with a similarly shaped piece of a healthy donor cornea.

96. A man whose blood group is not known meets with a serious accident and needs blood transfusion immediately. Which one of the blood groups mentioned below and readily available in the hospital will be safe for transfusion?

(a) O, Rh^- (b) O, Rh^+

(c) AB, Rh^- (d) AB, Rh^+

Ans. (a) : There is no A and B antigen on RBC of "O–"bloodgroup , So O, Rh– is known as universal blood donor.

97. "Metastasis" is the process by which:

(a) Cells divide rapidly under the influence of drugs

(b) Cancer cells spread through the blood or lymphatic system to other sites or organs

(c) The chromosomes in cell nuclei are attached to the spindle before moving to the anaphase poles

(d) Cancer cells are successfully inhibited to divide any further

Ans. (b) : Cancer cells spread through the blood or lymphatic system to other sites or organs.

"Metastasis" the spread of cancer cells from the place where they first formed to another part of the body. In metastasis, cancer cells break away from the original (primary) tumor, travel through the blood or lymph system, and form a new tumor in other organs or tissues of the body.

98. Match List I with List II and select the correct answer using the codes given below the lists:

	List I (Substance)		List II (Physiological role)
A.	Ptyalin	1.	Converts angiotensinogen in blood into angiotensin
B.	Pepsin	2.	Digests starch
C.	Renin	3.	Digests proteins
D.	Oxytocin	4.	Hydrolyses fasts
		5.	Induces contraction of smooth muscles

	A	B	C	D
(a)	2	3	1	5
(b)	3	4	2	5
(c)	2	3	5	1
(d)	3	1	2	1

Ans. (a) : The correct match is as follows:

	List-I (Substance)		List-II (Physiological role)
1.	Ptyaline	**A.**	Digests starch
2.	Pepsin	B.	Breakdown proteins into smaller peptides i.e. digest proteins
3.	Renin	C.	Convert angitensinogen in blood into angiotensis
4.	Oxytocin	D.	Induces contraction of smooth muscles.

99. Epiphytes are plants which depend on other plants for:

(a) food (b) mechanical support

(c) shade (d) water

Ans. (b) : They are an important source of food for many species. Typically, the older parts of a plant will have more epiphytes growing on them. Epiphytes differ from parasites in that they grow on other plants for physical support and do not necessarily affect the host negatively.

100. Antigen is a substance which:

(a) Destroys harmful bacteria

(b) Is used to treat poisoning

(c) Lowers body-temperature

(d) Stimulates formation of antibody

Ans. (d) : Antigens are glycoproteins found in RBCs. Due to this, there is a difference in human blood groups. On the basis of presence of antigen, four types of blood groups are found in humans - O, A, B, AB.

101. Which of the following features of DNA makes it uniquely suited to store and transmit genetic information from generation to generation?

(a) Complementarily of the two strands

(b) Double helix

(c) Number of base-pairs per turn

(d) Sugar-phosphate backbone

Ans. (b) : Double helix makes it uniquely suited to store and transmit genetic information from generation to generation.

102. Which of the following cell organelles play the most significant role in protein synthesis?

(a) Lysosome and Centrosome

(b) Endoplasmic reticulum and Ribosome

(c) Golgi apparatus and Mitochondria

(d) Lysosome and Mitochondria

Ans. (b) : The main function of endoplasmic reticulum is to synthesize all the fats and proteins that make up various membranes such as cell membranes, nucleus membranes, etc.

103. Assertion (A) : The boiling point of water decreases as the altitude increases.

Reason (R) : The atmospheric pressure increases with altitude.

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

Ans. (c) : The boiling point of water decreases as the altitude increase. As altitude increases atmospheric pressure decreases. Hence option (c) is correct.

104. Assertion (A) : Scientists can cut apart and paste together DNA molecules at will, regardless of the source of the molecules.

Reason (R) : DNA fragments can be manipulated using restriction endonucleases and DNA ligases

(a) Both A and R are individually true and R is the correct explanation of A

(b) Both A and R are individually true but R is NOT the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

Ans. (a) : The complementary properties of nucleotides of DNA are powerful tools for genetic engineering. DNA or any segment of a gene can be combined with other organisms' DNA using restriction endonuclease and DNA lipase. Such organisms that carry external DNA are called transgenic organisms.

ENVIRONMENT AND ECOLOGY

105. Consider the following statements regarding environmental issues of India:

1. Gulf of Mannar is one of the biosphere reserves.

2. The Ganga Action Plan, phase II has been merged with the National River Conservation Plan.

3. The National Museum of Natural History at New Delhi imparts non-formal education in environment and conservation.

4. Environmental Information System (ENVIS) acts as a decentralised information network for environmental information.

Which of these statements are correct?

(a) 1, 2 and 4 (b) 1, 2, 3 and 4

(c) 2 and 3 (d) 1, 3 and 4

Ans. (b) : Gulf of Mannar is one of India's 18 biosphere reserves declared so far. It is one of the world's richest region from a marine biodiversity perspective.

Ganga Action Plan Phase-II was merged with National River Conservation Plan in December, 1996.

National Museum of Natural History, established in 1972 and opened in 1978 was a museum located in New Delhi. The NMNH's mission was to promote environmental education. The museum was destroyed by fire on 26 April, 2016.

Environmental Information System (ENVIS) is a one stop, web enabled and comprehensive portal which provide information on environment and related subject areas.

106. Identify the correct order of the processes of soil-erosion from the following:

(a) Splash erosion, Sheet erosion, Rill erosion, Gully erosion

(b) Sheet erosion, Splash erosion, Gully erosion, Rill erosion

(c) Rill erosion, Gully erosion, Sheet erosion, Splash erosion

(d) Gully erosion, Rill erosion, Sheet erosion, Splash erosion

Ans. (a) : The correct order of soil erosion process is splash erosion, sheet erosion, rill erosion and gully erosion. Splash occurs when the raindrop hits the surface; explosive impact soil into individual particles. Sheet erosion is the detachment of soil particles by rain drop and their removal down slope by water flowing overland as a sheet in definite channels. Rill erosion refers to the development of small ephemeral concentrated flow paths which function as both sedimentary source and sediment delivery systems for erosion on hill slopes. Sheet erosion removes the this top layer by raindrop Rill erosion occurs in lines which are less than 30 cm deep. When these rills are deeper than 30 cm resulting channels are called as gully erosion. Gully erosion occurs when water flows in narrow channels after heavy rain and melting snow.

CURRENT AFFAIRS

107. The theme of Indian Science Congress, 2001 was:

(a) "Food, nutrition & environmental security"

(b) "Arrest declining interest in pure sciences"

(c) "Make India energy self-sufficient"

(d) Make India I.T. Superpower

Ans. (a) : The 88th Indian Science Congress was organised on 3-7 January 2001. The focus theme of this Congress was "Food nutrition and environment security". 108th Indian Science Congress from 3-7th January, 2022 at Symbiosis International University, Pune. Focal Theme - Science and Technology for "Sustainable Development with Women Empowerment."